Machine Learning Analysis of ME/CFS Phenotype Classification

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Project Overview & Key Contributions

ME/CFS Classification

Classify ME/CFS vs. healthy controls using survey data.

Symptom & Recovery Focus

Emphasis on patient-reported symptoms and recovery behaviors.

False-Negative-Averse

Model design prioritizes avoiding misclassifying ME/CFS patients.

Robust Dataset

Analyzed 1,196 records from 142 participants.

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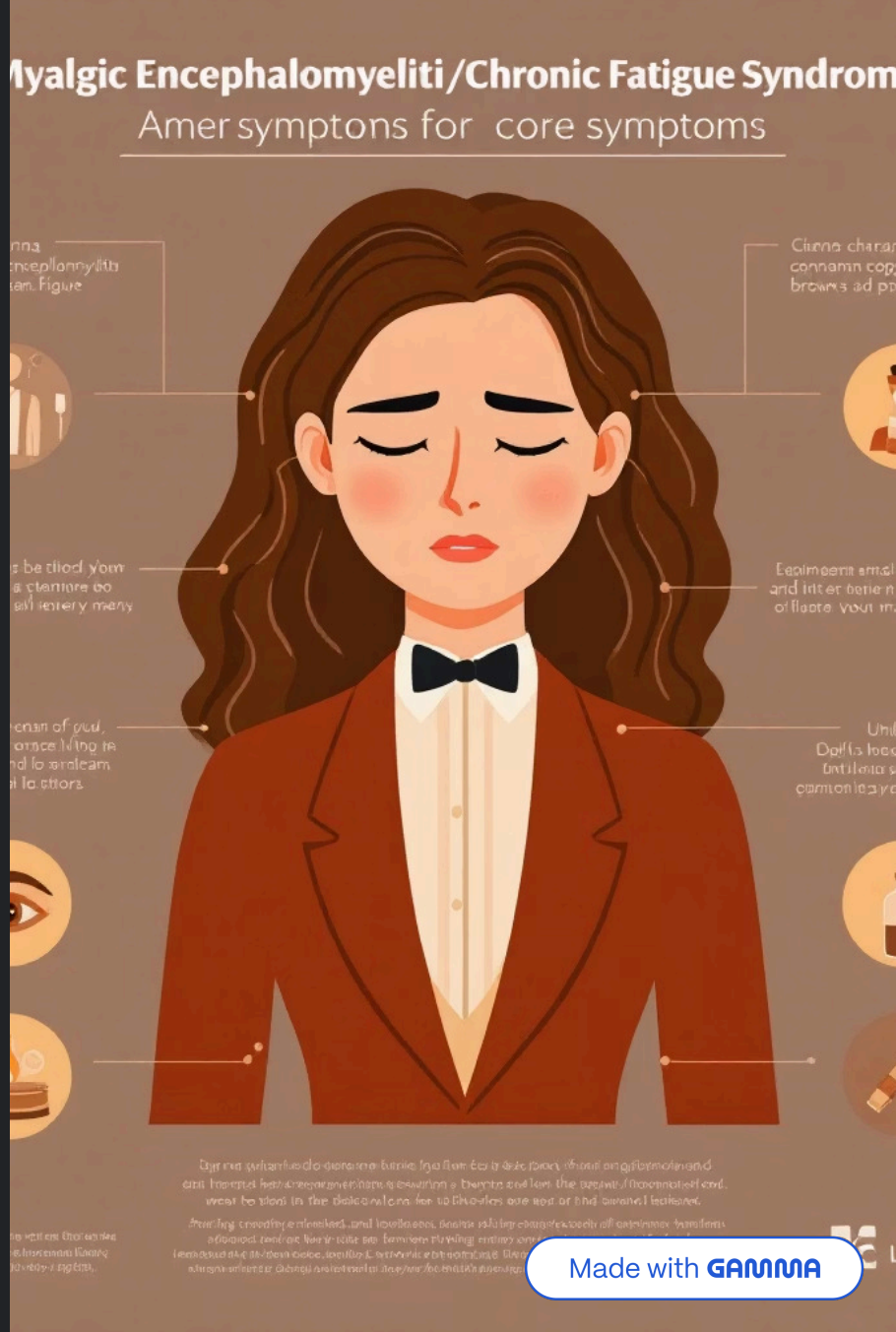
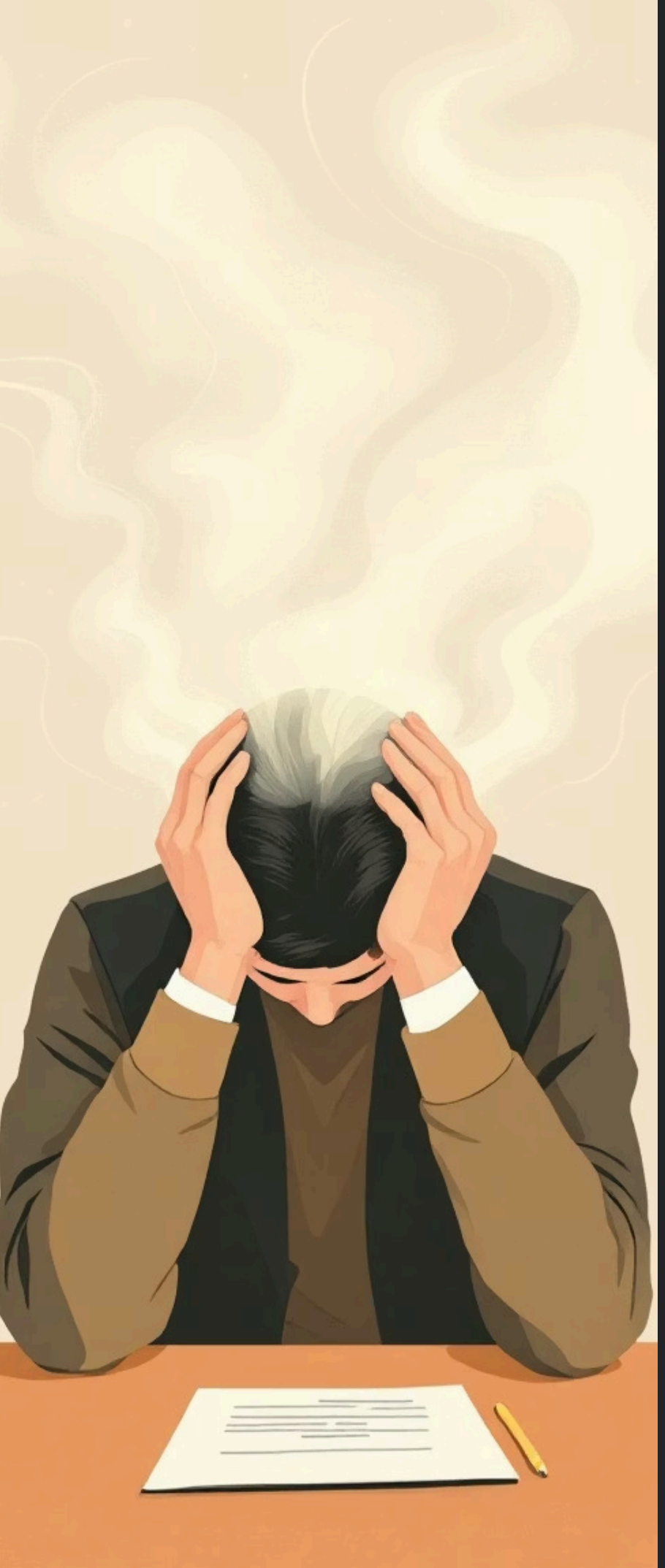
Last time / What is ME/CFS ?

- **Myalgic Encephalomyelitis / Chronic Fatigue Syndrome**
  A complex chronic illness affecting multiple body systems.
- **Core Symptoms:**
  Severe fatigue, cognitive impairment (“brain fog”), widespread pain, non-restorative sleep.
- **Post-Exertional Malaise (PEM):**
  Sudden worsening of symptoms after minimal physical or mental activity.
- **Prolonged Recovery:**
  Symptom recovery may take **days or weeks**, not hours.
- **Diagnostic Challenges:**
  No biomarker; symptoms overlap with other conditions → high misdiagnosis rates





ME/CFS – Why a Careful Model Matters

1

Chronic & Debilitating

Patients suffer from fatigue, brain fog, pain, and poor sleep.

2

Post-Exertional Malaise (PEM)

Severe crash after minimal activity, hindering daily life.

3

Prolonged Recovery

Recovery from PEM can take days or even weeks.

4

Diagnostic Challenges

Overlapping symptoms make accurate diagnosis difficult.

> False negatives (ME predicted as CTL) are medically more harmful than false positives.

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Dataset: survey_phenotype_export_for_rti.tsv

Key Dataset Characteristics

- 1,196 survey rows, 28 variables.
- 78 ME and 64 CTL participants.
- ~56% ME vs 44% CTL records.
- 7–14 surveys per participant (longitudinal).

Variables Overview

- Demographics: ParticipantID, Phenotype, AGE, GENDER, Study Site.
- Core Symptom Scores (0–10): Fatigue, Brain Fog, Sore Throat, Tender Nodes, Myalgia, Arthralgia, Headache, Sleep Disturbance, PEM.
- Recovery & Timing: Recovery Time (days), Survey Day, Day Sx.
- Summary/Diagnosis-like fields: Baseline Sx, Pretest Sx, SSS variants, MECFS duration.

ParticipantID	Sample_Source	GENDER	AGE	Phenotype	MECFS duration	Study Site	Sur
PI-055	NA	Female	44	CTL	ITH -1 1 3 0 0 0 5 1 3 0 Low 1.444444444		
PI-055	NA	Female	44	CTL	ITH 0 3 6 0 0 1 2 0 7 0 Low 2.111111111	Low 1	
PI-055	NA	Female	44	CTL	ITH 1 7 0 0 0 0 4 0 8 0 Low 2.111111111	1	
PI-055	NA	Female	44	CTL	ITH 2 1 0 0 0 0 1 0 0 0 Low 0.222222222	1	
PI-055	NA	Female	44	CTL	ITH 4 2 0 0 0 0 1 0 1 0 Low 0.444444444	1	
PI-055	NA	Female	44	CTL	ITH 6 0 0 0 0 0 0 0 0 0 Low 0	1	
PI-055	NA	Female	44	CTL	ITH 8 0 0 0 0 0 0 0 0 0 Low 0	1	
PI-055	NA	Female	44	CTL	ITH 10 0 0 0 0 0 0 0 0 0 Low 0	1	
PI-082	NA	Female	19	CTL	ITH -1 0 0 0 0 0 0 0 0 0 Low 0		
PI-082	NA	Female	19	CTL	ITH 0 0 0 0 0 0 0 0 0 0 Low 0	Low 1	
PI-082	NA	Female	19	CTL	ITH 1 0 0 0 0 0 0 0 0 0 Low 0	1	
PI-082	NA	Female	19	CTL	ITH 2 0 0 0 0 0 0 0 0 0 Low 0	1	
PI-082	NA	Female	19	CTL	ITH 4 0 0 0 0 0 0 0 0 0 Low 0	1	
PI-082	NA	Female	19	CTL	ITH 6 0 0 0 0 0 0 0 0 0 Low 0	1	
PI-082	NA	Female	19	CTL	ITH 8 0 0 0 0 0 0 0 0 0 Low 0	1	
PI-082	NA	Female	19	CTL	ITH 10 0 0 0 0 0 0 0 0 0 Low 0	1	
PI-153	NA	Female	44	CTL	ITH -1 2 2 2 0 0 0 0 2 1 Low 1		
PI-153	NA	Female	44	CTL	ITH 0 0 0 1 0 0 0 0 0 0 Low 0.111111111	Low 1	
PI-153	NA	Female	44	CTL	ITH 1 0 0 3 0 0 0 0 3 0 Low 0.666666667	1	
PI-153	NA	Female	44	CTL	ITH 2 0 0 2 0 0 0 0 3 0 Low 0.555555556	1	
PI-153	NA	Female	44	CTL	ITH 4 0 0 0 0 0 0 0 1 0 Low 0.111111111	1	
PI-153	NA	Female	44	CTL	ITH 6 0 0 0 0 0 0 0 0 0 Low 0	1	
PI-153	NA	Female	44	CTL	ITH 8 0 0 0 0 0 0 0 3 0 Low 0.333333333	1	
PI-153	NA	Female	44	CTL	ITH 10 0 0 0 0 0 0 5 2 0 Low 0.777777778	1	
PI-037	NA	Female	53	CTL	ITH -1 9 8 2 5 8 7 10 10 7 High 7.333333333		
PI-037	NA	Female	53	CTL	ITH 0 6 6 0 0 5 5 0 5 5 Mid 3.555555556	Int 2	
PI-037	NA	Female	53	CTL	ITH 1 9 8 1 7 1 6 8 7	2	
PI-037	NA	Female	53	CTL	ITH 2 7 6 0 0 5 3 3 6 5 Mid 3.888888889	2	
PI-037	NA	Female	53	CTL	ITH 4 6 5 0 0 1 1 2 6 0 Low 2.333333333	2	
PI-037	NA	Female	53	CTL	ITH 6 7 7 1 0 5 2 1 7 0 Mid 3.333333333	2	
PI-037	NA	Female	53	CTL	ITH 8 7 8 0 0 6 1 2 6 0 Mid 3.333333333	2	
PI-037	NA	Female	53	CTL	ITH 10 8 7 0 0 8 1 4 6 0 Mid 3.777777778	2	
PI-037	NA	Female	53	CTL	ITH 12 6 7 0 0 6 1 1 6	2	

snapshot of the original dataset

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Cleaning and Preventing Label Leakage

01

Column Elimination

Drop empty or non-predictive columns (e.g., Sample_Source).

02

Diagnosis Field Removal

Exclude diagnosis-like summary fields (e.g., Baseline Sx, Pretest Sx, SSS variants, MECFS duration).

03

ME Duration Treatment

Treated as structural missingness, then excluded from feature matrix X.

04

Feature Matrix Exclusion

Phenotype, ParticipantID, and MECFS duration excluded from feature matrix X to ensure model conservatism.

By avoiding label leakage, the achieved zero-FN result is meaningful and not an artifact of data processing.

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Imputation and Feature Encoding

Participant-Wise Median Fill

Key symptoms (Fatigue, Brain Fog, PEM, etc.) filled using individual participant medians.

KNNImputer

k=5 for remaining numeric NaNs, ensuring no missing values in the final dataset.

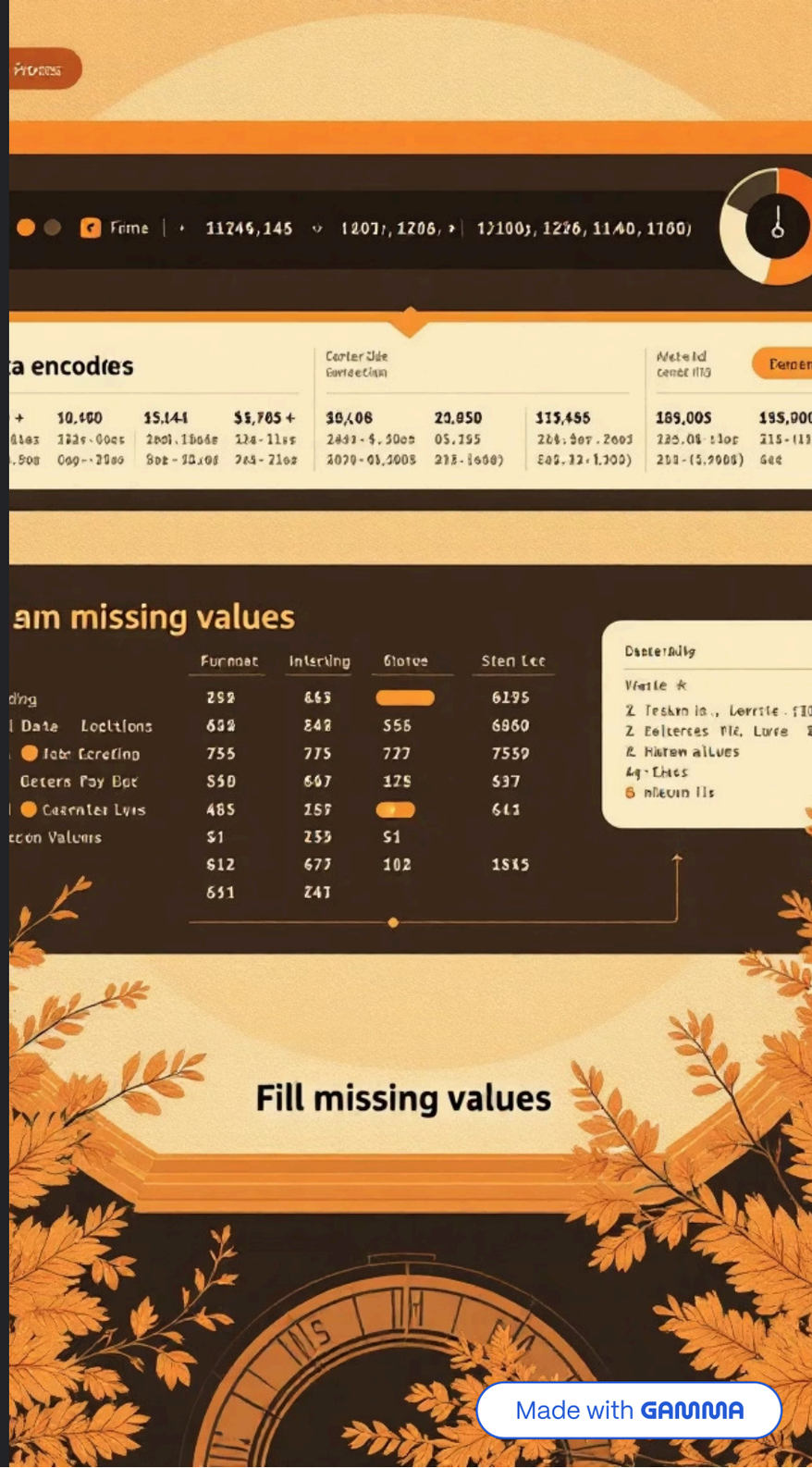
One-Hot Encoding

Applied to categorical variables like GENDER and Day Sx.

Final Feature Matrix (X)

Numeric, dense, and leakage-controlled, ready for model training.

Robust preprocessing is crucial for reliable estimates of recall, F-beta, and permutation importance.



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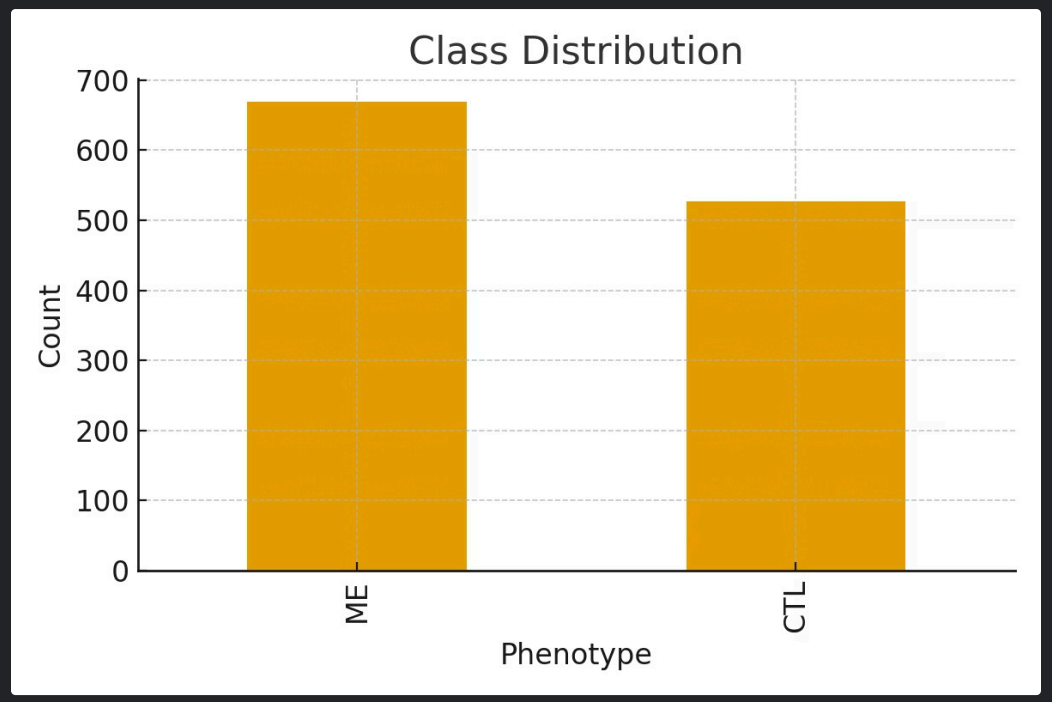
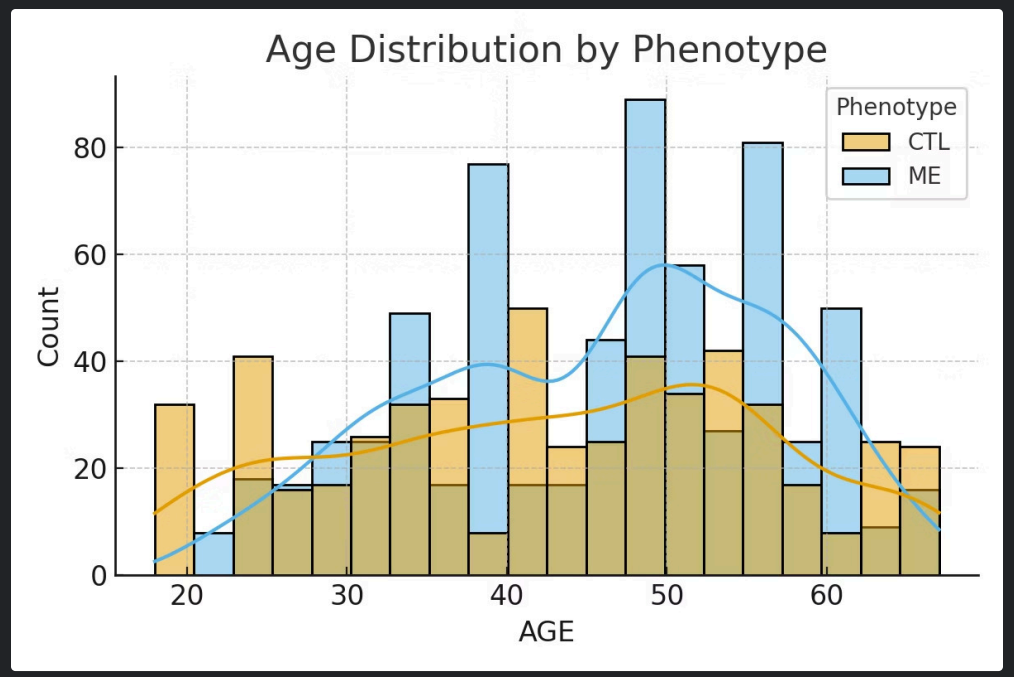
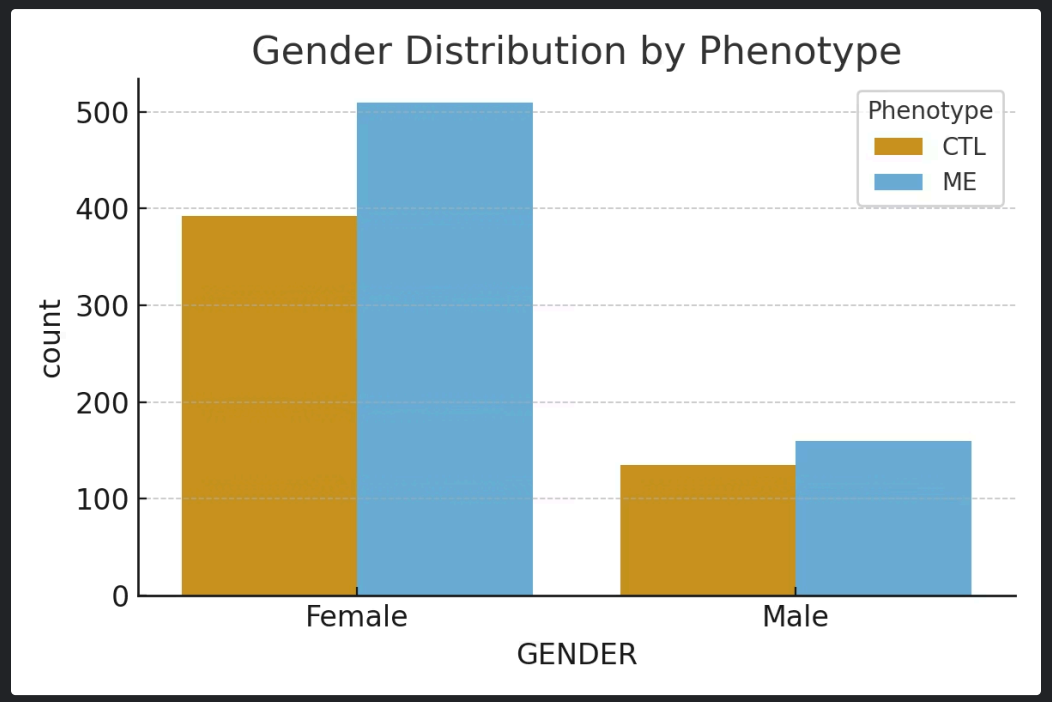
EDA – Classes, Age, Gender, Recovery

Dataset Snapshot

- Records: 669 ME vs 527 CTL (reasonably balanced).
- Age: Overlapping distributions; ME slightly older (overall age ≈ 44.7 ± 12.2).
- Gender: ~75% female in both groups (~901 F / 295 M total).
- Recovery Time: ME much slower than CTL.







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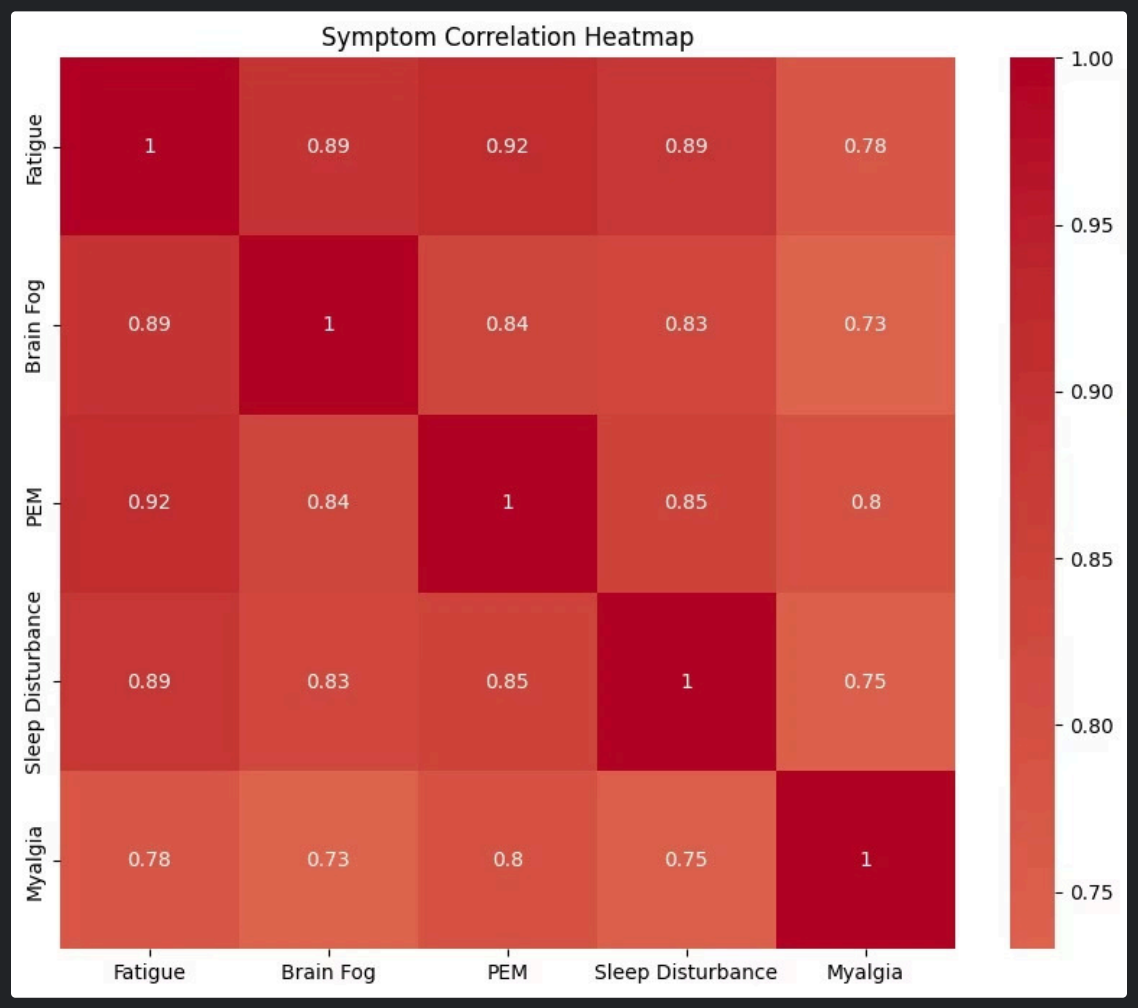
EDA – Symptom Severity & Correlations

Symptom Severity

CTL: most symptoms near 0–1; ME: symptoms cluster around 6–7+ (0–10 scale). E.g., ME fatigue ≈ 6.8 vs CTL ≈ 0.8.

Strong Correlations

Core symptoms are highly correlated (e.g., fatigue–PEM: 0.73–0.92), motivating feature engineering.



The significant gap in symptom severity between ME and CTL facilitates reliable ME detection, reducing false negatives.

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Feature Engineering & Scenarios

1

PEM_x_Recovery

PEM score × Recovery Time.

2

Neuro_Cluster

Mean of Fatigue, Brain Fog, Sleep Disturbance.

3

Physical_Cluster

Mean of Myalgia, Arthralgia, Tender Nodes.

4

Total_Symptom_Score

Sum of selected symptom scores.

Feature Set Cases

- **CASE 1:** Original features only (raw symptom/demographic info).
- **CASE 2:** Engineered features only (compact, clinically meaningful set).

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Models, Cross-Validation & Best Hyperparameters

GroupShuffleSplit by ParticipantID (20% test) ensures robust evaluation across patients.

model	case 1	case 2
RF	0.983	0.929
XGB	0.986	0.936
SVM	0.983	0.947
GBC	0.987	0.927
KNN	0.954	0.944
LR	0.976	0.944

GBC with original features yielded the highest CV accuracy (0.987), selected as the final model.

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Final Model, False Negatives & Key Features

Final GBC Model

Tuned GradientBoostingClassifier

High Performance

Test: Accuracy 0.968, ROC AUC 1.000.

Zero ME→CTL Errors

Confusion matrix: [[106, 8], [0, 133]] — perfect recall for ME.

F2 and F1.5 Scores

Emphasize recall (F2 ≈ 0.988, F1.5 ≈ 0.982), confirming false-negative aversion.

Permutation Importance

Recovery & PEM×Recovery highlighted with 95% CIs.

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Final Model: Gradient Boosting Classifier (GBC)

Model Performance

- **Accuracy:** 0.968
- **ROC AUC:** 1.000
- **Recall (ME class):** 1.00 → *zero ME→CTL false negatives*
- **F2 Score:** 0.988
- **F1.5 Score:** 0.982

Confusion Matrix (Test Set):
[[106, 8],
[0, 133]]

Predicted CTL	Predicted ME
106	8
0	133

```
'GBC': (GradientBoostingClassifier(), {
        'n_estimators': [100, 200],
        'learning_rate': [0.05, 0.1, 0.2],
        'max_depth': [3, 4, 5]
    }),
```

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Confidence Intervals: TRUE Features vs Engineered Features

Original Features

Recovery Time: 0.161 ± 0.028
Pretest Sx_Int: 0.128 ± 0.030
Baseline Sx_Mid: 0.043 ± 0.020
PEM: 0.013 ± 0.012
Brain Fog: 0.002 ± 0.004

Engineered Features

PEM_x_Recovery: 0.270 ± 0.044
Total_Symptom_Score: 0.153 ± 0.039
Physical_Cluster: 0.002 ± 0.005
Neuro_Cluster: 0.001 ± 0.006

Analysis of permutation importance, including 95% confidence intervals, reveals the significant predictive power of both original and engineered features. Notably, the engineered feature **PEM_x_Recovery** demonstrates the highest importance score, highlighting its strong contribution to distinguishing between ME/CFS and control groups. Among original features, **Recovery Time** and **Pretest Sx_Int** are consistently important. The larger importance values for engineered features underscore the benefit of feature engineering in creating more discriminative variables for classification.

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Future work

- **External Validation:**
  Evaluate the model on independent ME/CFS cohorts to test generalizability.
- **Longitudinal Modeling:**
  Use temporal ML models (e.g., LSTMs, time-series analysis) to capture symptom changes across surveys.
- **Threshold Optimization:**
  Tune decision thresholds to further minimize false negatives without large increases in false positives.
- **Cost-Sensitive Learning:**
  Assign higher penalties to ME→CTL errors during training to reinforce recall.
- **Data Expansion:**
  Integrate wearable sensor data (heart rate, sleep tracking) for objective ME/CFS markers.
- **Explainability Tools:**
  Apply SHAP, ICE, and feature attribution techniques for clinician-friendly interpretations

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thank you

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